MACHINE LEARNING MODEL OPTIMIZATION & PERFORMANCE COMPARISON

Project: California Housing Price Prediction

Deliverable: Task 2 Technical Report

1. Executive Summary

The primary objective of this project is to develop an end-to-end regression pipeline to estimate house prices using the California Housing Dataset. The workflow incorporates feature standardization, multi-algorithm training, structured metric evaluation (Root Mean Squared Error and R^2 Score), and visual residual inspection. Among the evaluated models—Linear Regression, Ridge Regression, and Decision Tree Regressor—the Decision Tree Regressor ($depth=5$) demonstrated the highest predictive capability, producing an RMSE of 0.7397 and an R^2 score of 0.5825.

2. Methodology & Implementation Pipeline

- **Dataset Overview:** The California Housing dataset consists of 20,640 instances with 8 numeric predictive attributes: Median Income, Housing Median Age, Average Rooms, Average Bedrooms, Population, Average Occupancy, Latitude, and Longitude. The target variable is House Price (measured in units of $100,000).
- **Feature Standardization:** Features exist across vastly different numerical ranges. StandardScaler was applied across input features to center data around a zero mean and unit variance, preventing gradient instability and ensuring distance-sensitive algorithms do not over-weight high-magnitude attributes.
- **Data Partitioning:** Standardized features were split using an 80/20 train-test ratio (test_size=0.2, random_state=42) to ensure that all model evaluations reflect generalized performance on completely unseen data.
- **Algorithm Selection:**
    - *Linear Regression:* Serves as the primary parametric baseline.
    - *Ridge Regression ($\alpha=1.0$):* Employs L_2 regularization to reduce overfitting and penalize large feature coefficients.
    - *Decision Tree Regressor ($max\_depth=5$):* A non-parametric partition model with explicit depth limits to capture non-linear relationships without overfitting.

3. Experimental Results & Discussion

Models were trained on the training subset and evaluated against the identical held-out test subset:

Model	**RMSE**	**R2 Score**	**Key Observations**
Linear Regression	0.7456	0.5758	Provides an adequate linear baseline.
Ridge Regression	0.7456	0.5758	Yields identical metrics to OLS at $\alpha=1.0$ due to low multicollinearity penalty demand.
Decision Tree Regressor	0.7397	0.5825	Achieves lowest error and highest variance explanation via non-linear feature splits.

Note: With targets denominated in units of \$100,000, an RMSE of ~0.74 corresponds to an average prediction deviation of ~\$74,000.

4. Visual Validation & Error Analysis

A scatter plot comparing actual versus predicted values against a 45-degree reference line illustrates key model behaviors:

- **Linear Constraints:** Linear models under-predict high-tier properties due to rigid planar assumptions.
- **Ceiling Cap Phenomenon:** The dataset imposes an upper cap on valuations at \$500,000 (\$5.0). The Decision Tree stabilizes near boundary conditions more effectively than unbounded linear models by grouping high-density boundary clusters into dedicated leaf nodes.

5. Final Model Selection & Conclusion

The **Decision Tree Regressor ($max\_depth=5$)** is selected as the best-performing model. It reduces prediction error (RMSE: 0.7397) and explains higher variance (R^2: 0.5825) compared to linear alternatives. The final model artifact was serialized to disk using joblib (best_housing_model.pkl) for production reproducibility.